

Sensation

CT

Troubleshooting Guide

WCS (water/water)

Troubleshooting Guide WCS (water/water)

This document is valid for:
SOMATOM Sensation 10/16/Cardiac
SOMATOM Sensation 40/64/Cardiac 64/Open

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Safety information

Read the following instructions as well as the most recent version of

- **General Safety Notes** (TD00-000.860.01.xx.xx)
- **Product-specific safety Information** (CT02-023.860.01.xx.xx)

before trouble shooting the WCS.

The following instructions and the above mentioned **Safety documentation** contain important safety information as well as information about tools to be used.

NOTE

Only use the <Read> button in the functions <Table load/modify>. All other functions are not necessary for service, and are normally for development staff only.

=> Improper handling can result in system malfunctions.

NOTE

Operating the WCS water pump without water (e.g. during filling/refilling the gantry) will destroy the pump.

=> Do not operate the water pump without water!

NOTE

Do not operate the system in customer mode at the increased water pressure as used for the test. Always return the water pressure to the nominal operating pressure of 2.0 bar $\pm$ 0.2 bar, with the heat exchanger switched off.

Tools

- A standard tool kit
- 2 water hoses with 1/2" connections for filling and draining the system
- Manometer (included with delivery of the WCS)
- Absorbent cloth
- Blotting paper

Overview

The Water Cooling system and its components has to ensure the proper temperature inside the gantry. Therefore, errors caused by a defective WCS (or insufficient cold water supply of the hospital) can generate:

- temperature error messages for components within the gantry
- temperature error messages for the MCU
- image quality problems

This section provides additional troubleshooting information, in the event that information in either the detailed message or functional descriptions is insufficient.

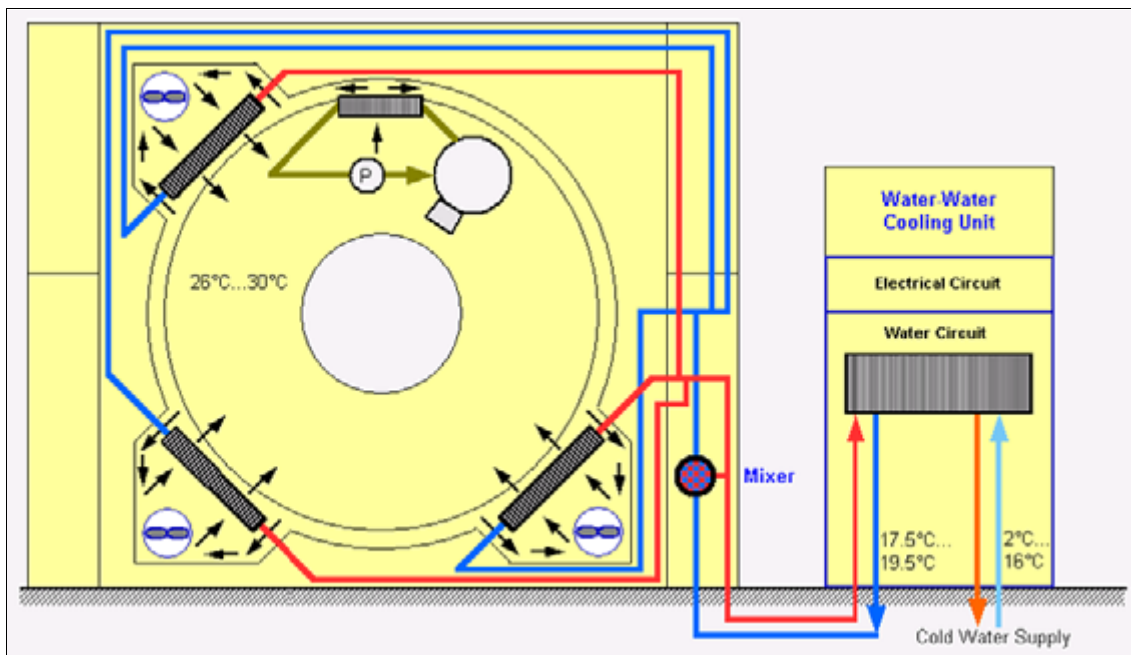


Fig. 1: Cooling circuit overview

Temperature value

Nominal temperature value

Internal gantry temperature

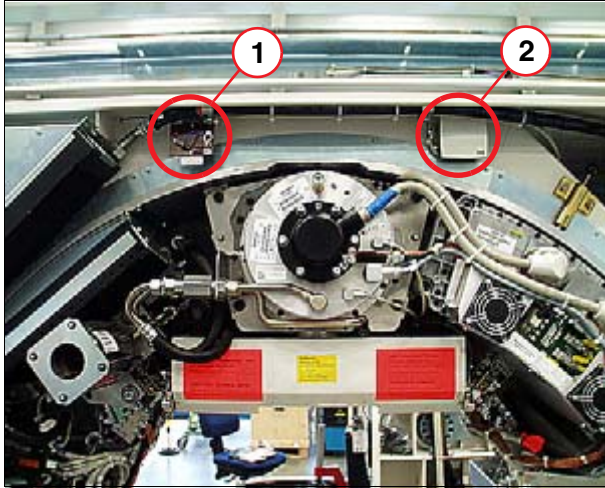


Fig. 2: Temperature Sensor in Gantry

Pos. 1 B 305

Pos. 2 B 302

- ⇒ B302 = the current temperature value, used for regulation.
B305 = the safety temperature switch. In case of overheating ($> 43^{\circ}\text{C}$), the PDC switches off the power to the PSS.
- ⇒ Nominal temperature:
 $26^{\circ}\text{C} \pm 4^{\circ}\text{C}$
- ⇒ Switching values
B302 $> 35^{\circ}\text{C}$ will generate a "Warning" to the user. (and Eventlog)
B305 $> 43^{\circ}\text{C}$ will switch off the Gantry completely. After the temperature is below $< 40^{\circ}\text{C}$, the Gantry can be switched on again.

Water temperature inside the WCS

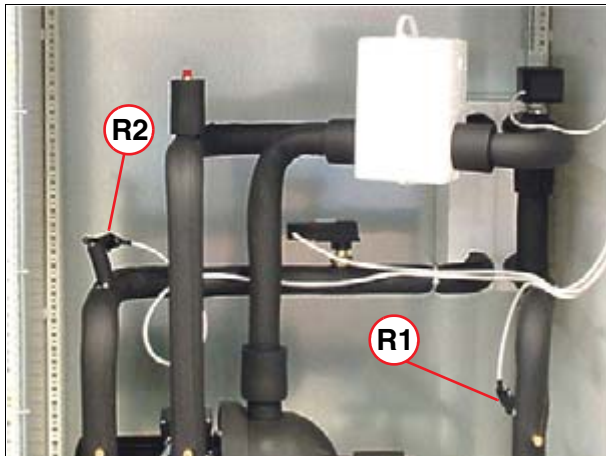


Fig. 3: Temperature Sensor in WCS

- ⇒ Temperature Sensor in the WCS (used for regulation inside WCS):
R1 = the mixed water temperature to gantry (Secondary water circuit).
R2 = the cold water temperature from hospital water circuit (Primary cooling circuit).
- ⇒ Nominal temperature:
R1 = 17.5°C to 19.5°C.
R2 = 2°C to 16°C.

Current temperature values

The current temperature values of the various sensors can be evaluated via the syngo service software.

How to get the current temperature values

1. From the user interface, select **Options -> Local Service**
2. In the authentication page, type in the **Service Password** and Click **OK**
3. In the Home menu, click the **Control** button
4. Click **Table load/modify**
 - ⇒ The following page appears.
5. You can access the control system table by

- ➡ Click **GO**, and the following page appears

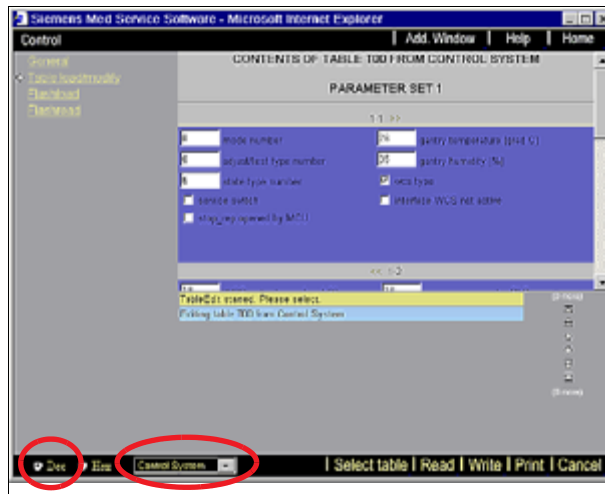


Fig. 4: Example of Parameter Set 1

6. Select **Dec**, and click **Read**

- ⇒ The corresponding table will be activated.
 Gantry temp. (B302) = “**gantry temperature**”
 Water temp. in primary cooling circuit (R2)= “**primary cooling media temp**”
 Water temp. in secondary cooling circuit (R1) = “**WCS water temp.**”

NOTE

Only use the <Read> button of the functions <Table load/modify>. All other functions are not necessary for service, and are normally for development staff only.

=> Improper handling can result in system malfunctions.

NOTE

This table T00 include the current status of N1 WCS controller. Please refer (Tab. 3 / p. 15).

If current temperature is out of tolerance

Gantry temperature

There is possibility as below

- Primary cooling circuit problems
 - ⇒ Refer (How to get the current temperature values / p. 8)
- Secondary cooling circuit problems
 - ⇒ Refer (How to get the current temperature values / p. 8)

- Cooling fan in Gantry problems
 - ⇒ Check/replace the fan for the heat exchanger as described in replacement instruction
CT02-023-841.01.XX.XX
- Temperature Sensor (B302) problems
 - ⇒ check/replace the temperature sensor.

Primary cooling circuit

There is possibility as below

- Primary cooling circuit problems
 - ⇒ Contact the person in charge of the primary cooling circuit.
- Temperature Sensor (R2) problems
 - ⇒ Check/replace the temperature sensor as described in replacement instruction. CT02-023-841.05.XX.XX. As a reference, the resistance of the sensors will be appr. 2000 Ohms measured at 15°C.

Secondary cooling circuit

There is possibility as below

- Primary cooling circuit problems
 - ⇒ Refer ([How to get the current temperature values / p. 8](#))
- WCS problems
 - ⇒ Refer ([Electrical circuit / p. 11](#))
- Water leaks
 - ⇒ Refer ([Water circuits / p. 16](#))
- Temperature Sensor (R1) problems
 - ⇒ Check/replace the temperature sensor as described in replacement instruction. CT02-023-841.05.XX.XX. As a reference, the resistance of the sensors will be appr. 2000 Ohms measured at 15°C.

Electrical circuit

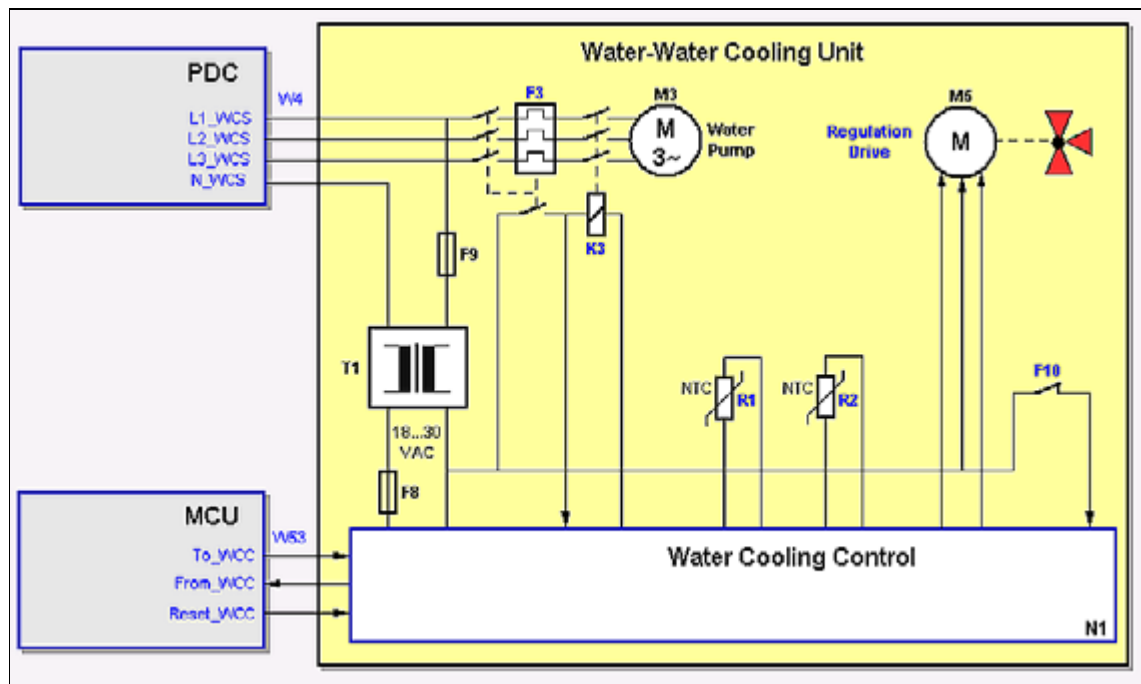


Fig. 5: WCS water/water Electrical circuit overview

Please refer to the latest version of the Sensation 16 /Sensation Cariat Functional description (page 51201). Both sensors, R1 and R2, are PTC-type! As a reference, the resistance of the sensors will be approx. 2000 Ohms measured at 15°C.

The figure above will be updated soon.

N1 - WCS controller

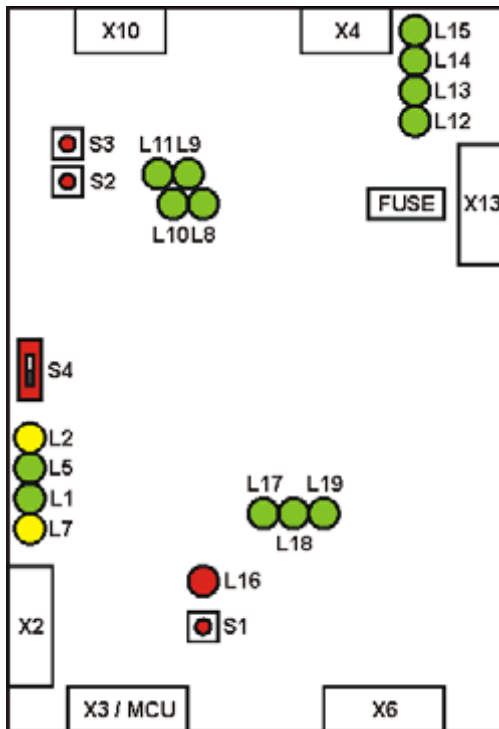


Fig. 6: N1 - WCS Controller (schematic layout)

For further investigations within the WCS, it is necessary to know the current status of the N1 WCS controller. You can obtain the current status from LED on N1 ([LED"s on the N1 controller / p. 12](#)) or the system control table via syngo service software ([The system control table / p. 13](#)).

LED"s on the N1 controller

Tab. 1 LED"s on the N1 controller

LED / color	Normal status	Description for WCS (water/water)
L1 - green	blinking	Program active
L2 - yellow	off	Service active
L3 - n.a.	n.a.	not mounted
L4 - n.a.	n.a.	not mounted
L5 - green	on, if -->	Communication active (telegram traffic)
L6 - n.a	n.a.	not mounted
L7 - yellow	off	if on --> malfunction
L8 - green	n.a.	not used
L9 - green	n.a.	not used
L10 - green	n.a.	not used

LED / color	Normal status	Description for WCS (water/water)
L11 - green	n.a.	not used
L12 - green	on, if -->	50Hz or 60 Hz is available
L13 - green	n.a.	not used
L14 - green	on	Cooling pump protection F3 / OK off, if --> over current
L15 - green	on	Water flow (Primary cooling circuit) / OK
L16 - red	off	on, if --> MCU reset occurs
L17 - green	on, if -->	Mixing circuit: colder
L18 - green	on, if -->	Mixing circuit: warmer
L19 - green	on	Cooling pump running
L20 - green	n.a.	not used
L21 - n.a.	n.a.	not mounted
L22 - n.a.	n.a.	not mounted
L23 - n.a.	n.a.	not mounted
L24 - n.a.	n.a.	not mounted

The system control table

1. From user interface, select **Options -> Local Service**
2. In the authentication page, type in the **Service Password** and Click **OK**
3. In the Home menu, click **Control** button
4. Click **Table load/modify**

➡ The following page appears.

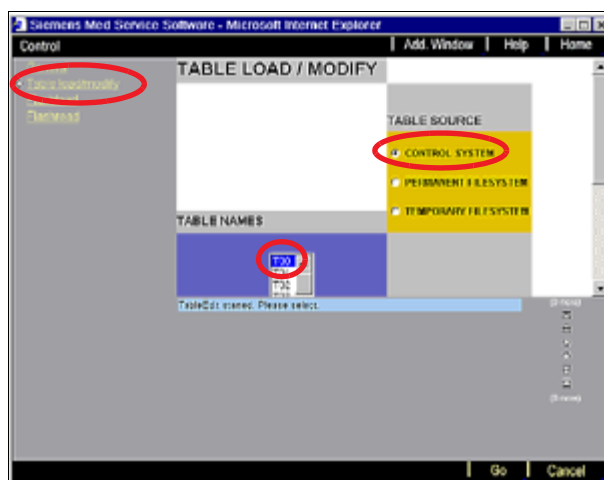


Fig. 7: Table load / modify page

5. You can access the control system table by
- Selecting **T00** for TABLE NAME
 - Selecting **CONTROL SYSTEM** for TABLE SOURCE
- ⇒ Click **GO**, and the following page appears

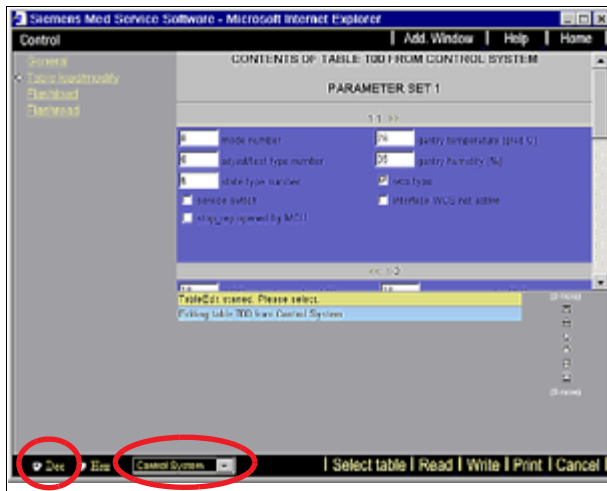


Fig. 8: Example of Parameter Set 1

6. Select **Dec**, and click **Read**
- ⇒ The corresponding table will be activated.

NOTE

Only use the <Read> button of the functions <Table load/modify>. All other functions are not necessary for service, and are normally for development staff only.

=> Improper handling can result in system malfunctions.

Tab. 2 T00 table contents

	Normal condition
gantry temperature	current temperature inside gantry
gantry humidity	current humidity inside gantry
WCS water temp.	current temperature in secondary cooling circuit (R1)
compressor 1 active	Off / Only for W/A
compressor 2 active	Off / Only for W/A
ventilator Y/D active	Off / Only for W/A
cooling pump active	On
cooling med. out of tol.	Off

	Normal condition
cooling wat. out of tol.	Off
cooling sys. out of tol.	Off
cooling sys. defective	Off
no water flow	Off
mot protector pump	Off
mot protector ventilator	No / Only for W/A
primary cooling media temp	current temperature in primary cooling circuit (R2)

NOTE

This table T00 include the current temperature value. Please check the tolerance as described in [\(Temperature value / p. 7\)](#).

Other components parts in electrical circuit

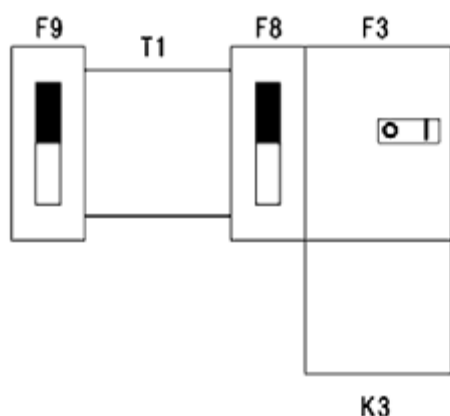


Fig. 9: Other components parts

Tab. 3 Fuse, Measurement points and LED's

	Description
F3	Cooling pump over current protection
F8	Circuit breaker 3A
F9	Circuit breaker 1A
K3	Cooling pump contactor
T1	Transformer

Water circuits

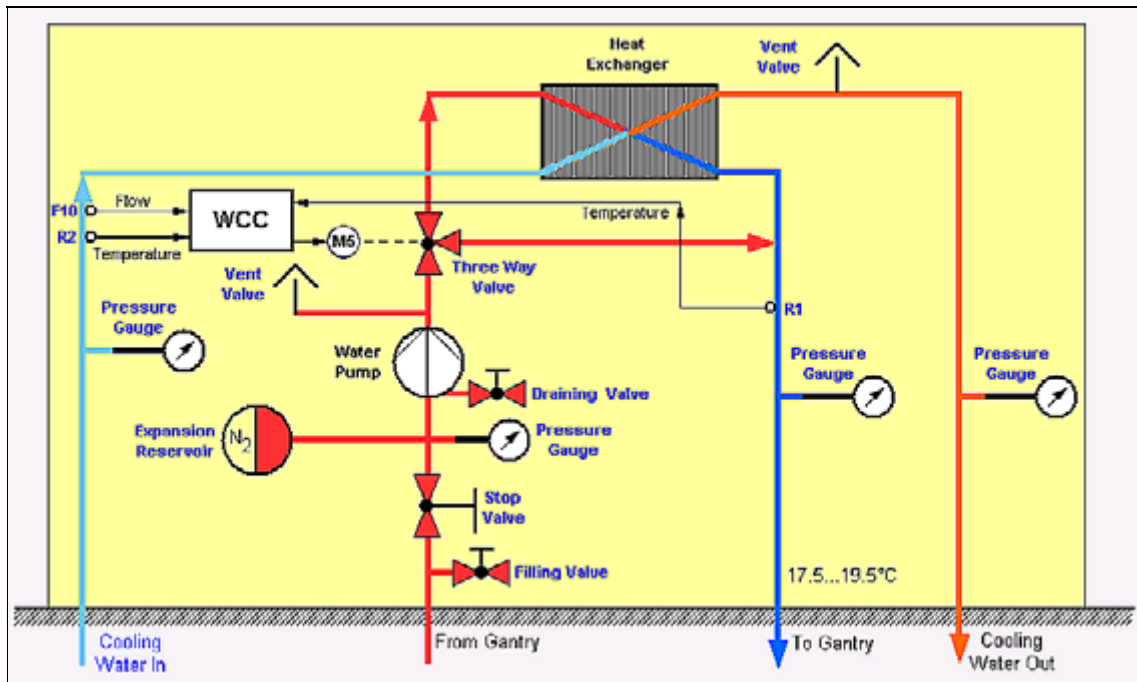


Fig. 10: WCS Water circuit overview

Draining and filling the cooling system

NOTE

Operating the WCS water pump without water (e.g. during filling/refilling the gantry) will destroy the pump.

=> Do not operate the water pump without water!

Testing for leaks requires increase or decrease the water pressure.

Draining process to decrease the water pressure

1. Switch off the WCS via breaker F14 in PDC.
2. Remove the shutter of the filling and draining valves (4/Fig. 11 / p. 17) and connect the water hoses .
3. Place absorbent cloth under the filling and draining valves, because water may escape.
4. Route the water hoses to a syphon or a bucket.

5. The red shut-off valve for the gantry cooling system must be open.
 - ⇒ The lever is lined up with the connection pipe (1/ Fig. 11 / p. 17).

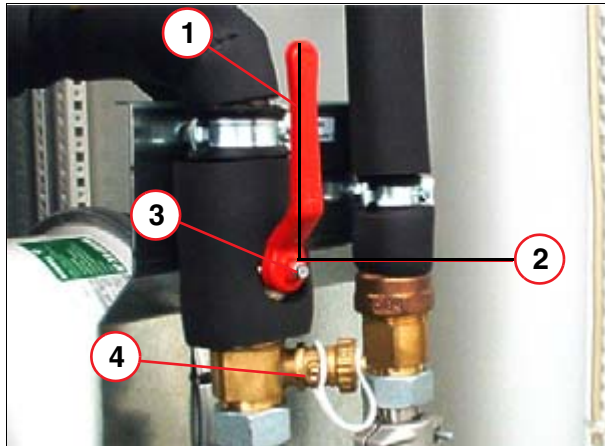


Fig. 11: Shut-off valve at WCS "Gantry water inlet"

Pos. 1	Open
Pos. 2	Close
Pos. 3	Shut-off valve
Pos. 4	filling and draining valve

6. Connect the monometer at WCS" Gantry water outlet".
7. Open the filling and draining valve by turning it 90° counterclockwise and using the valve cap. Watch the manometer!
 - ⇒ The valve is open when the notch on the valve is lined up with the connection pipe.
8. Drain until the water pressure reaches 2.0 bar $\pm$ 0.2 bar and close the filling and draining valve by turning it 90° clockwise using the valve cap.
9. Switch on the WCS.

NOTE

After WCS is running for a while, please check that the water pressure is 2.0 bar $\pm$ 0.2 bar.

Filling the cooling system

Please always refer to the latest version of the document

Installation Instructions CT02-023.814.01.xx.xx. Filling the cooling system.

Reasons for a drop in pressure

After installation

Drop in pressure once or twice after start-up. After the system is filled a maximum of 3 times, pressure should remain constant. The pressure should not drop by more than 1.0 bar in 6 months.

Possible causes:

- Excessive amounts of air in the cooling system.
- Warm water was used to fill the system (loss in volume when cooled to 18°C).
- The hose for filling the system was not filled with water when connected to the WCS.
- Widening of water hoses within the first weeks after system installation.

After the system has been filled several times

The drop in pressure is >1.0 bar in 6 months. The minimum value of the static pressure is 1.0 bar.

Possible causes:

- Screw connections are not water-tight.
- Leaks in the affected system components (e.g., in the water pump, 3-way valve, Spirovent, expansion reservoir, heat exchanger, connection WCS <--> Gantry, within gantry). Continue with [\(Testing for leaks with WCS switched off / p. 18\)](#) or [\(Testing for leaks in individual WCS components / p. 19\)](#).

Testing for leaks with WCS switched off

NOTE

Do not operate the system in customer mode at the increased water pressure as used for the test. Always return the water pressure to the nominal operating pressure of 2.0 bar $\pm$ 0.2 bar, with the heat exchanger switched off.

=> Please refer to [\(Draining process to decrease the water pressure / p. 16\)](#).

NOTE

Do not perform scans at this time.

1. Switch off the WCS.
2. Connect the filling hose to the cooling system.
 - ⇒ Fill the hose with water prior to connecting it to the system to avoid introducing air into the system. Any air in the filling hose will later lead to a loss of pressure.
3. Fill it until the pressure reaches a maximum (< 7.5 bar) as described in [\(Filling the cooling system / p. 17\)](#).

NOTE

The maximum pressure depends on the water supplier and the location of the site. In case the pressure is not able to reach 7.5 bar, test will be performed with your maximum.

4. Record the water pressure in the WCS:
 - ⇒ If the pressure drops by more than 10% within 1 hour, there is a leak in the system.

5. Check the WCS, the gantry, and the full length of all the water hoses for possible leaks (use blotting paper).
6. If leaks are found, replace the correspond parts. Please refer the each replacement instruction.

Gantry / Replacement of the Parts CT02-023.841.01.xx.xx

WCS / Replacement of the Parts CT-023.841.05.xx.xx

7. After the replacement of the parts, make sure the leaks are not found with this test.
8. If leaks are not found with this test, the individual components can be tested.

Testing for leaks in individual WCS components

If leaks are not found with the previous test using increased pressure, individual components can be tested as follows.

3-way valve in Gantry right stand and WCS

Overview - location of the 3-way valves



Fig. 12: 3-Way Valve in Gantry stand

How to:

1. Remove the black insulation covering the 3-way valve
2. Place absorbent cloth under the 3-way valve, because water may escape during test.

3. Open/close the both 3-way valve approx. 10 times. The valve can be opened and closed by turning the white screw of the adjustment drive CW (open) or CCW (closed) with a 5mm allen key ([1/Fig. 13 / p. 20](#)).



Fig. 13: Example of Adjustment drive

4. Check the 3-way valve for leaks (use blotting paper).
5. If leaks are found, replace the correspond 3-way valve. Please refer the each replacement instruction.
 - 3-Way valve in Gantry stand CT02-023.841.01.xx.xx
 - 3-Way valve in WCS CT-023.841.05.xx.xx
6. After the replacement of the 3-way valve, make sure the leaks are not found with this test.
7. Drain until the water pressure reaches 2.0 ± 0.2 bar as described in ([Draining process to decrease the water pressure / p. 16](#)).
8. Put the black insulation covering the 3-way valve

NOTE

Do not operate the system in customer mode at the increased water pressure as used for the test. Always return the water pressure to the nominal operating pressure of 2.0 bar ± 0.2 bar, with the heat exchanger switched off.

=> Please refer to ([Draining process to decrease the water pressure / p. 16](#)).

Cooling pump

Overview - location of the devices



Fig. 14: Cooling pump

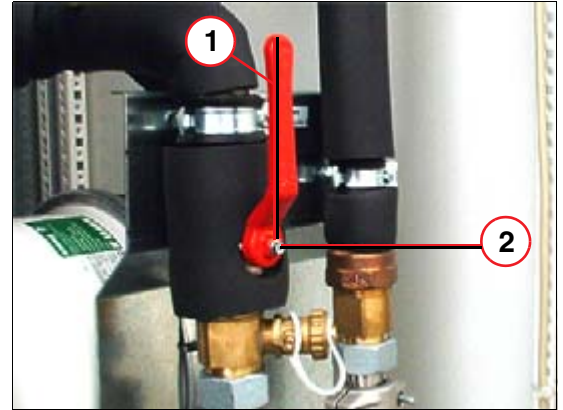


Fig. 15: Shut-off valve

NOTE

Do not perform the following test with the increased water pressure. Even short-term test with the increased water pressure will destroy the water pump. Ensure the water pressure is at the nominal operating pressure of 2.0 bar $\pm$ 0.2 bar with the WCS switched off.

=> Please refer to [\(Draining process to decrease the water pressure / p. 16\)](#).

How to:

1. Shift the black insulation sleeves away from the nuts.
2. For approx. 5 minutes, close the shut-off valve ([2/ Fig. 15 / p. 21](#)) of the cooling pump being tested. This forces the pump to work under increased pressure.
3. Check the pump for leaks (use blotting paper).
4. If leaks are found, replace the cooling pump as described in replacement instruction.
Replacement of Parts CT-023.841.05.xx.xx
5. After the replacement of the cooling pump, make sure the leaks are not found with this test.

Additional hints

Pressure Value

- The nominal water pressure should be 2.0 bar $\pm$ 0.2 bar with the WCS switched OFF.
- When the WCS is switched off (pump is off), pressure at the water inlet (in front of the pump) and at the water outlet (behind the pump) are equal.
- When the WCS is switched on (pump is on), a relative pressure difference of approx. 2.5 bar can be measured in front and behind the pump.

- The absolute pressure measured via a manometer depends on the length of the hose and the type of operation (50/60 Hz).

Anti-freeze agent

- Only to the primary (hospital) cooling circuit can up to 35% anti-freeze agent be added. The necessary water flow and water temperature have to be corrected. A diagram is available. Please refer to the latest version of the Planning Guide (CT02-023.891.01.XX.XX).

NOTE

Anti-freeze agents reduce the cooling ability of water and anti-freeze mixture.

- It is prohibited to add an anti-freeze agent to the secondary (gantry) cooling circuit.

Others

- Excessive amounts of air in the cooling system (after filling) can damage the pump!
- To operate an additional external water pump or valve, a potential-free contact (max. 1 Amp, 220 Volts) is available in the PDC - PDS.

Chapter	Change	Reason
FMP	SOMATOM 40 was added	136660 (P30G)

